

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of three strategies: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure strategy includes the explicit zero-gap nine-point theorem and the nonzero-gap residual witness sets. Appendices collect the exact local reach calculations certifying that the relevant finite sets do not fit in a unit C triangle. The paper is organized as one proof, and the two mixed support-arc intersections are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

CONTENTS

1. Introduction and Proof Outline	1
2. Common Geometry and Structural Reductions	6
3. Trace-Length Mechanism	8
4. Area-Loss Estimate	11
5. Finite-Enclosure Obstructions	13
6. Completion of the Proof	28
Appendix A. Structural, Shared-Local, and Signed-Center Optimization	29
Appendix B. Trace-Length Optimization	33
Appendix C. Area-Loss Optimization	35
Appendix D. Nonzero-Gap Finite-Enclosure Optimization	35
Appendix E. Zero-Gap Nine-Point Optimization	42
Appendix F. Exact Unequal-Radius Support-Arc Intersections	48

1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

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and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

Then

$$(1) \quad \begin{aligned} \mathcal{H}^1(\partial H) &= 6, \\ \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) &= 6, \\ \mathcal{H}^1(S) &= 12. \end{aligned}$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

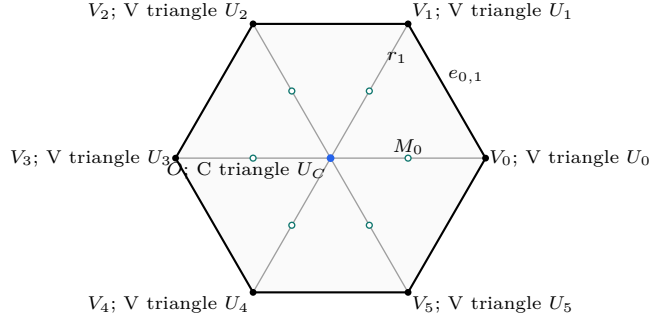


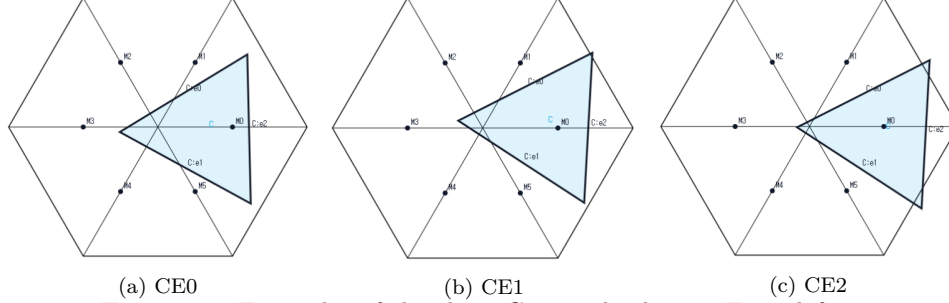
FIGURE 1. The center, vertices, radial segments, and radial mid-points of H .

1.2. Finite classifications. Set

$$N_E(T_C) := \left| \left\{ i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0 \right\} \right|.$$

The only possible values are 0, 1, 2. We call the corresponding C triangle types CE0, CE1, CE2; thus

$$\begin{aligned} T_C \text{ has type CE0} &\iff N_E(T_C) = 0, \\ T_C \text{ has type CE1} &\iff N_E(T_C) = 1, \\ T_C \text{ has type CE2} &\iff N_E(T_C) = 2. \end{aligned}$$



(a) CE0 (b) CE1 (c) CE2
 FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
(1, 0)	Vd0
(2, 0)	Vd0
(1, 1)	Vd1
(1, 2)	Vd2
(2, 1)	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$

$$B_i := \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$

$$C_i := \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}.$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i . The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$(2) \quad N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase reaches are used only after a selection satisfying

$$0 \leq a_i \leq A_i,$$

$$0 \leq b_i \leq B_i,$$

$$0 \leq c_i \leq C_i.$$

They never define N_+ .

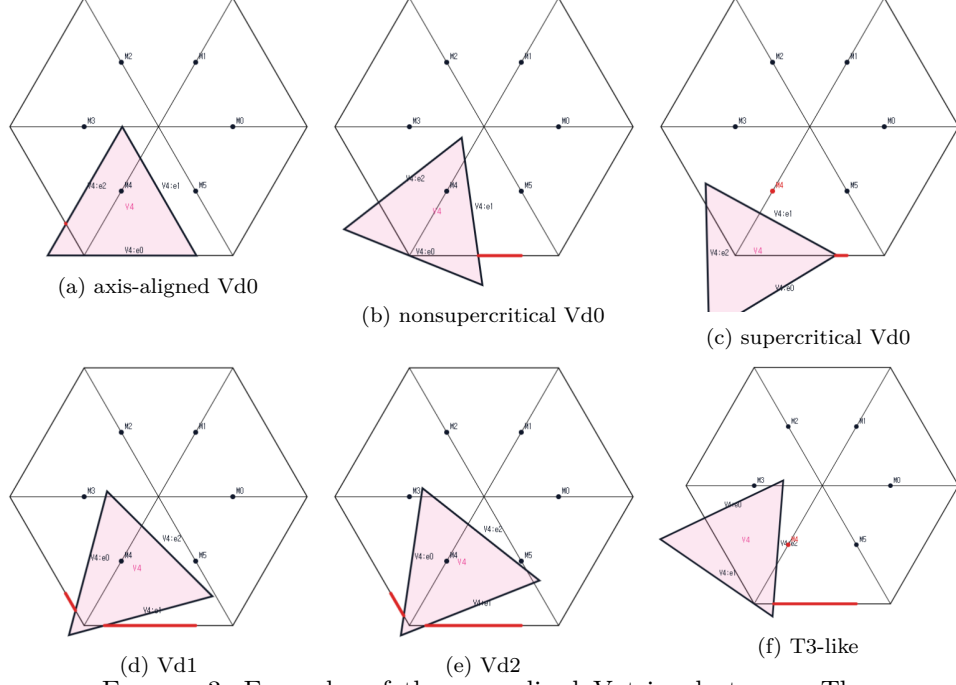


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

The incident open traces are

$$\begin{aligned} U_i \cap e_{i,i+1} &= X_i([0, B_i)), \\ U_{i+1} \cap e_{i,i+1} &= X_i((1 - A_{i+1}, 1]). \end{aligned}$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})).$$

The displayed trace identities give

$$J_i \neq \emptyset \iff B_i + A_{i+1} \leq 1.$$

Equality gives a singleton gap, which remains uncovered. Put

$$N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|.$$

For the case tables, put

$$\begin{aligned} d &:= |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|, \\ t &:= |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2, 1)\}|, \end{aligned}$$

and

$$N_{\text{sp}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : n(T_i) > 0\}| = d + t.$$

The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Lemma 2.9 proves the fundamental equivalence

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is the unique route for a hypothetical cover. The full classification proof is Proposition 2.14.

N_{gap}	N_+	d	t	$N_E(T_C)$	mechanism
0	0	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	Trace length
0	1	$d \geq 1$	$t \geq 0$	$N_E(T_C) \in \{0, 1, 2\}$	Trace length
0	1	$d = 0$	$t \geq 1$	$N_E(T_C) \in \{0, 1, 2\}$	Area loss
0	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{0, 1, 2\}$	Finite enclosure
0	$N_+ \geq 2$	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	Area loss
$N_{\text{gap}} \geq 1$	$N_+ \geq 2$	$d + t \leq 6$		$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\text{gap}} \geq 1$	$N_+ \in \{0, 1\}$	$N_+ + d + t \geq 3$		$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\text{gap}} \geq 1$	0	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\text{gap}} \geq 1$	0	$d \geq 1$	$d + t \leq 2$	$N_E(T_C) \in \{1, 2\}$	Trace length
$N_{\text{gap}} \geq 1$	0	$d = 0$	$t \in \{1, 2\}$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\text{gap}} \geq 1$	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\text{gap}} \geq 1$	1	$d = 0$	$t = 1$	$N_E(T_C) \in \{1, 2\}$	Finite enclosure
$N_{\text{gap}} \geq 1$	1	$d = 1$	$t = 0$	$N_E(T_C) = 1$	Trace length
$N_{\text{gap}} \geq 1$	1	$d = 1$	$t = 0$	$N_E(T_C) = 2$	Trace and finite enclosure

TABLE 1. The exhaustive routing. Strategy 1 is a trace-length argument, Strategy 2 sums area loss, and Strategy 3 forces a finite residual set into the C triangle and proves an equilateral-enclosure obstruction. Nonzero gap count forces center class CE1 or CE2. In that part the high-count rows are applied first; the final two rows separate the CE1 length route from the detailed CE2 finite-enclosure placement assembly.

1.4. Organization of the proof. Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the finite residual-hull principle and the nonzero-gap and zero-gap enclosure

obstructions. Section 6 then applies the zero-gap matrix before the nonzero-gap rows of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. COMMON GEOMETRY AND STRUCTURAL REDUCTIONS

The introduction fixes the exact C triangle and V triangle classifications, the actual maximal reaches A_i, B_i, C_i , the selected lower bounds a_i, b_i, c_i , and the boundary gaps J_i . This section proves that those definitions give the finite structural alternatives used by all three mechanisms. Coordinate evaluations needed in the proofs are collected in Appendix A.

2.1. Open and closed formulations.

Proposition 2.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side 1.

Lemma 2.2 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

2.2. C triangle and V triangle classifications. Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.3 (Exhaustive center types). *The closed C triangle T_C has exactly one of the types CE0, CE1, CE2; equivalently,*

$$N_E(T_C) \leq 2.$$

Lemma 2.4 (Local wedge). *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H ; if both adjacent radial traces have positive length, at least two vertices belong to H .*

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.5 (Exact-trace normalization of raw $(3, 0)$ roles). *Let U be an original open V triangle at V_i , put $T = \bar{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\hat{U}$, with closure $\hat{T}$, such that*

$$(3) \quad V_i \in \hat{U}, \quad (o(\hat{T}), n(\hat{T})) = (2, 0),$$

and

$$(4) \quad T \cap H = \hat{T} \cap H, \quad U \cap H = \hat{U} \cap H.$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.6 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like in the exact dictionary of the introduction.*

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.7 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V triangle at V_i . There is a same-orientation translate $\hat{T}$ such that*

$$V_i \in \text{relint}(a \text{ side of } \hat{T}), \quad T \cap H \subsetneq \hat{T} \cap H,$$

and $(o(\hat{T}), n(\hat{T})) = (2, 1)$, while

$$\text{int}(T) \cap (H \setminus \{V_i\}) \subseteq \text{int}(\hat{T}).$$

Since V_i becomes a boundary point, $\hat{T}$ is only a closed-trace majorant.

2.3. Initial reaches and boundary gaps. The exact reach definitions are those in the introduction. The following elementary fact records how their closed maxima and open suprema are used.

Lemma 2.8 (Initial reach segments). *On the three segments from V_i toward V_{i-1}, V_{i+1}, O , the closed parameter sets are respectively*

$$[0, A_i], \quad [0, B_i], \quad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \quad [0, B_i), \quad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

The gap J_i is the exact set difference defined in the introduction. The name includes the equality case $B_i + A_{i+1} = 1$: then J_i is a singleton missed by both incident open V triangles.

Lemma 2.9 (Exhaustiveness of the gap split). *Every J_i is empty or one closed interval, with singleton intervals retained. Under a hypothetical cover,*

$$N_{\text{gap}} = |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|,$$

and every nonempty J_i lies in a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 permits respectively at most 0, 1, 2 gaps, and

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Lemma 2.10 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(5) \quad A_i + B_i \leq 1.$$

It is therefore not counted by N_+ .

2.4. Strict handoffs and midpoint forcing.

Proposition 2.11 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i . If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Lemma 2.12 (Self-midpoint obstruction). *Let u, v be unit vectors meeting at 120° . If a closed unit equilateral triangle contains*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2},$$

then $a + b \leq 1$. Consequently

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proposition 2.13 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and T_C is CE1 or CE2, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 , but O is not interior to it.

2.5. Exhaustive structural routing.

Proposition 2.14 (Exhaustive structural reduction). *Every hypothetical cover admits the distinct C triangle and V triangles used in Table 1, and its actual maximal reaches determine exactly one row of that table.*

3. TRACE-LENGTH MECHANISM

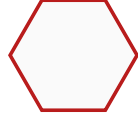
For a closed triangle T and a rectifiable target E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Passing from an original open triangle to its closure can only enlarge its trace. Hence an open cover of E implies

$$(6) \quad \mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^5 L_E(T_i).$$

The mechanism applies this inequality only to the perimeter ∂H or the full skeleton S , under the hypotheses stated below.

 ∂H : length 6 S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton interfaces.

Lemma 3.1 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Lemma 3.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.3 (No-support skeleton cap). *If*

$$A_i + B_i \leq 1,$$

$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \quad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Lemma 3.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.5 (Positive-support rescuer). *Suppose $N_E(T_C) \in \{1, 2\}$ and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Theorem 3.6 (Direct CE1/CE2 skeleton count). *If*

$$N_E(T_C) \in \{1, 2\},$$

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

and

$$(7) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

3.2. Boundary trace register. The exact evaluation is in Appendix B; the following table is the complete interface used by the routing argument.

Theorem 3.7 (Boundary trace table). *For closures of original open triangles, the following bounds hold.*

	<i>exact hypothesis</i>	<i>established class</i>	<i>exact bound</i>
	$N_E(T_C) = 0$	CE0	$L_{\partial H}(T_C) = 0$
	$N_E(T_C) = 1$	CE1	$L_{\partial H}(T_C) \leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
	$N_E(T_C) = 2$	CE2	$L_{\partial H}(T_C) < \frac{1}{2}$
(8)	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i \leq 1$	Vd0	$L_{\partial H}(T_i) \leq 1$
	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i > 1$	Vd0	$L_{\partial H}(T_i) \leq \frac{2}{\sqrt{3}}$
	$(o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}$	Vd1 or Vd2	$L_{\partial H}(T_i) < \frac{1}{2}$
	$(o(T_i), n(T_i)) = (2, 1)$	T3-like	$L_{\partial H}(T_i) < 1.$

For every V triangle,

$$(9) \quad L_{\partial H}(T_i) = A_i + B_i.$$

Corollary 3.8 (Branches closed by the common budgets). *The following exact alternatives are impossible.*

- (1) $N_E(T_C) = 0$, $N_+ = 1$, and $d \geq 1$;
- (2) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (3) $N_E(T_C) = 1$, $N_+ = 1$, and $d \geq 1$;
- (4) $N_E(T_C) = 2$, $N_+ = 1$, and $d \geq 2$;
- (5) $N_E(T_C) = 2$, $N_+ = 1$, and there is an index i such that

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset;$$

- (6) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (7) $N_E(T_C) \in \{1, 2\}$ and $N_+ \geq 2$.

Proposition 3.9 (Vd2 adjacent-midpoint branch). *Assume*

$$N_E(T_C) = 2, \quad N_+ = 1, \quad (d, t) = (1, 0).$$

If the unique non-Vd0 triangle T_i satisfies

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then ∂H cannot be covered.

3.3. Routing conclusion.

Proposition 3.10 (Branches closed by the trace-length mechanism). *The following routing entries and hybrid subcase are impossible.*

- (1) $N_{\text{gap}} = 0$ and $N_+ = 0$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, and $d \geq 1$;
- (3) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (5) $N_E(T_C) = 1$, $N_+ = 1$, and $(d, t) = (1, 0)$;
- (6) the subcase of $N_E(T_C) = 2$, $N_+ = 1$, and $(d, t) = (1, 0)$ in which the unique non-Vd0 triangle T_i satisfies

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset.$$

4. AREA-LOSS ESTIMATE

The area-loss method records how much of each V triangle must lie outside H . The local estimate applies to every V triangle; the sharper T3-like estimate uses its classified geometry. A final cyclic argument adds these losses without imposing a center type.

Put $A_\Delta = \sqrt{3}/4$ and normalize all areas by A_Δ . At V_0 define

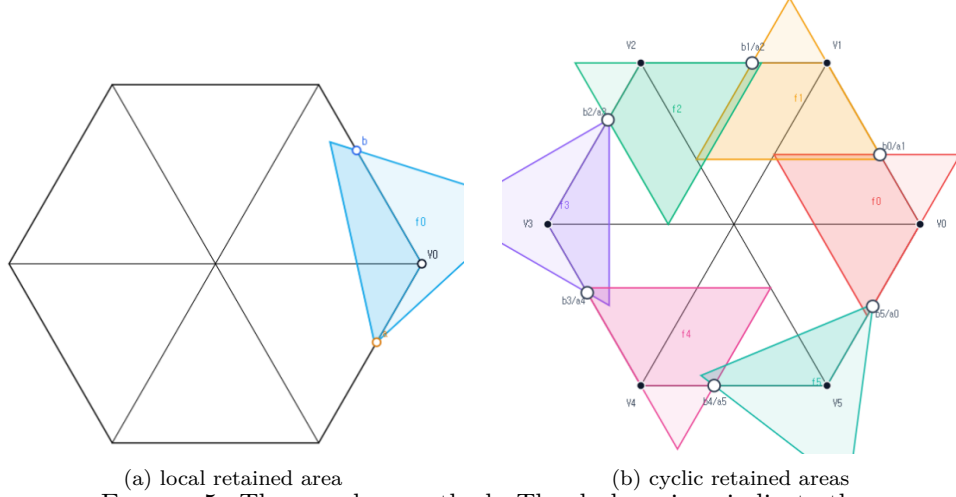
$$\mathcal{T}(a, b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\},$$

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}, \quad \mathcal{L}(a, b) := \inf_{T \in \mathcal{T}(a, b)} \mathcal{L}(T),$$

where $a, b \in [0, 1]$ and $\mathcal{T}(a, b) \neq \emptyset$. Hexagon symmetry gives the corresponding family at every V_i . In particular, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i,$$

then the rotated copy of T_i belongs to $\mathcal{T}(a_i, b_i)$.



(a) local retained area (b) cyclic retained areas
 FIGURE 5. The area-loss method. The dark regions indicate the portions retained in H ; the exact quantities are $\mathcal{L}(T)$ and $\mathcal{L}(a, b)$.

4.1. Local area loss.

Theorem 4.1 (Local square loss). *For every feasible (a, b) ,*

$$(10) \quad \mathcal{L}(a, b) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(11) \quad \mathcal{L}(a, b) \geq \max\{a, b\}^2.$$

The same bounds hold for every $T \in \mathcal{T}(a, b)$.

4.2. T3-like loss.

Theorem 4.2 (Direct T3-like loss). *Let T_i be a V triangle with*

$$(o(T_i), n(T_i)) = (2, 1),$$

and let

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i.$$

Then

$$(12) \quad a_i + b_i \leq 1.$$

Moreover, if

$$0 \leq \mu \leq \frac{1}{2}, \quad a_i, b_i \geq \mu,$$

then

$$(13) \quad \mathcal{L}(T_i) \geq 2\mu - 4\mu^2.$$

4.3. Cyclic accumulation. Use the strict handoffs from theorem 2.11:

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i.$$

They satisfy

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

Lemma 4.3 (Cyclic area-loss certificate). *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

They also satisfy

$$\left. \begin{aligned} &|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| = 1, \\ &\forall i \in \mathbb{Z}/6\mathbb{Z}, \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0), (2, 1)\}, \\ &\exists j \in \mathbb{Z}/6\mathbb{Z}, \quad (o(T_j), n(T_j)) = (2, 1) \end{aligned} \right\} \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proposition 4.4 (Branches closed by area). *The following branches are impossible:*

- (1) $N_{\text{gap}} = 0$ and $N_+ \geq 2$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, $d = 0$, and $t \geq 1$.

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subseteq \mathbb{R}^2$, define

$$\Lambda(K) := \inf \left\{ s > 0 : \begin{array}{l} \exists T \text{ [} T \text{ is a closed equilateral triangle of side } s, \\ K \subseteq T \end{array} \right\}.$$

Thus $\Lambda(K)$ is the least equilateral enclosure side for K .

Lemma 5.1 (Compact set in an open unit triangle). *If K is compact, U is an open unit equilateral triangle, and $K \subset U$, then*

$$\Lambda(K) < 1.$$

5.1. Four-anchor and neighboring-arm capacities. Normalize at V_0 and define the four-anchor hull

$$\begin{aligned} K(a, b, c) &:= \text{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\}, \\ \mathcal{A} &:= \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}. \end{aligned}$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

The exact admissible set, attainment, and the evaluation of these extrema are proved in theorem A.9.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T \text{ [} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a), \quad 0 \leq a, b \leq 1, \quad a + b \leq 1.$$

Their coordinate evaluation is deferred to theorem D.6.

Let selected lower bounds satisfy

$$0 \leq a_j \leq A_j, \quad 0 \leq b_j \leq B_j \quad (0 \leq j \leq 5),$$

$$a_j^2 + a_j b_j + b_j^2 \leq 1 \quad (0 \leq j \leq 5).$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\mathcal{H}^1(T_{i-1} \cap r_i) > 0 \implies a_{i-1} + b_{i-1} \leq 1,$$

$$\mathcal{H}^1(T_{i+1} \cap r_i) > 0 \implies a_{i+1} + b_{i+1} \leq 1.$$

Define

$$\Gamma_i := \max \left(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \right. \\ \left. \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\} \right), \quad D_i := (1 - \Gamma_i)V_i.$$

The points D_i are the radial witnesses used below.

Lemma 5.2 (Type-aware radial forcing). *For every i ,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Lemma 5.3 (Common-pair domination). *If*

$$p, q \geq 0, \quad p + q \leq 1,$$

then

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Lemma 5.4 (Open trace endpoint). *Let U be an open equilateral triangle and $T = \overline{U}$. If $T \cap [V, O]$ is the closed interval from V to P , then $P \notin U$. If a noncentral point $Q \in [V, O]$ lies in U , then $\mathcal{H}^1(U \cap [V, O]) > 0$.*

5.2. Actual gaps and the case register. For an actual gap on $e_{i,i+1}$, the definition from theorem 2.9 gives exactly

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

This remains a closed singleton when $B_i = 1 - A_{i+1}$. Equivalently,

$$U_i \cap e_{i,i+1} = X_i([0, B_i]), \quad U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Thus every point of J_i , including its endpoints, is center-forced under the skeleton-cover hypothesis.

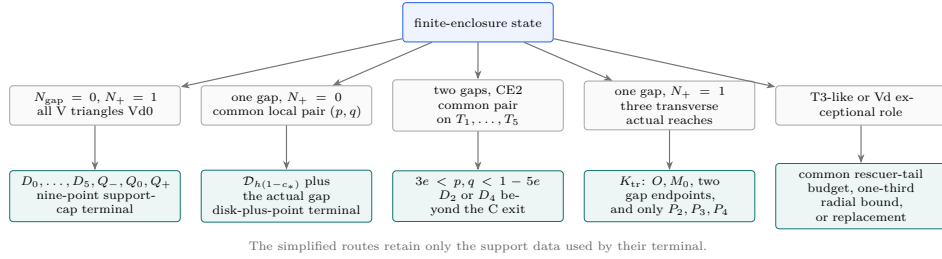


FIGURE 6. Roadmap of the simplified finite-enclosure chapter. The one-gap all-Vd0 witness uses only three transverse radial endpoints; the two-gap terminal uses the interval $3e < p, q < 1 - 5e$; and the T3-like and Vd1 paths share one rescuer-tail lemma.

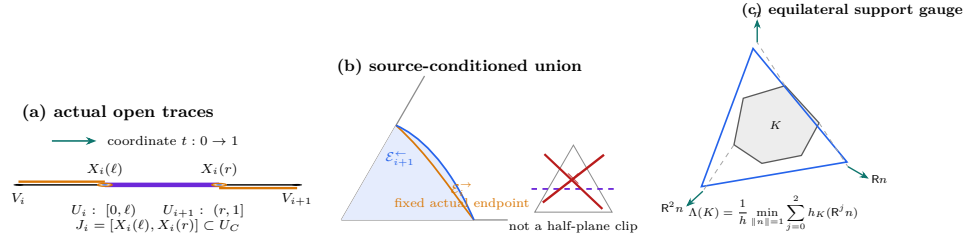


FIGURE 7. Three conventions used throughout the finite-enclosure argument. (a) A hollow outer marker records nonmembership in an open V trace, while the filled center marker records selection of the same endpoint as a center-forced witness. (b) A trace-exact envelope is obtained by restricting the source family to the prescribed actual maximal reach and then taking its union; it is not an affine half-plane clipping. (c) Equilateral enclosure is measured by three support directions separated by 120° . Incidences and interval orders are exact; lengths are schematic.

For the one-Vd rows define the normalized indices by

$$\{\sigma\} := \{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\},$$

$$\{\tau\} := \{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}.$$

Rows A–E and the replacement router assume $S \subseteq U_C \cup \bigcup_{i=0}^5 U_i$; row F assumes the hypothetical cover of H .

Table 2: Exact finite-enclosure case register. The symbolic columns determine the row; the final column gives its concise geometric route.

gap conditions	actual reaches	role and placement	route
$N_{\text{gap}} = 1$ $J = J_0$ $J_0 = X_0([\ell, r])$ $0 < \ell \leq r < 1$	$N_+ = 0$	$d = 0$ $t \in \{0, 1, 2\}$ $N_E(T_C) \in \{1, 2\}$	A. Complementary gap.
$N_{\text{gap}} = 2$	$N_+ \in \{0, 1\}$ $N_+ + t \leq 2$ $N_+ t = 0$	$d = 0$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	
$N_{\text{gap}} = 1$ $J = J_0$ $J_0 = X_0([\ell, r])$ $0 < \ell \leq r < 1$	$N_+ = 1$ $A_0 + B_0 > 1$	$(d, t) = (0, 0)$ $N_E(T_C) \in \{1, 2\}$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	C. Transverse seven-point witness.
$N_{\text{gap}} \in \{1, 2\}$	$N_+ = 1$	$(d, t) = (0, 1)$ $(o(T_0), n(T_0)) = (2, 1)$ $N_E(T_C) \in \{1, 2\}$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$ $(d, t) = (1, 0)$	
$N_{\text{gap}} \in \{1, 2\}$	$N_+ = 1$ $A_0 + B_0 < 1/2$	$(o(T_0), n(T_0)) = (1, 1)$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$ $(d, t) = (1, 0)$ $\sigma = 0$	D_V. Vd1 supported tail.
$N_{\text{gap}} \in \{1, 2\}$	$N_+ = 1$	$\tau \in \{1, 5\}$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$ $(d, t) = (1, 0)$ $\sigma = 0$	
$N_{\text{gap}} \in \{1, 2\}$	$N_+ = 1$	$\tau \in \{2, 3, 4\}$ $N_E(T_C) = 2$ $T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$ $(d, t) = (0, 0)$	E_n. Nonadjacent radial separation.
$N_{\text{gap}} = 0$	$N_+ = 1$	$H \subseteq U_C \cup \bigcup_{i=0}^5 U_i$	
			F. Zero-gap nine-point obstruction.

gap conditions	actual reaches	role and placement	route
		$(d, t) = (1, 0)$	
		$\sigma \in \{1, 2, 3, 4, 5\}$	
		$\tau \in \{1, 2, 3, 4, 5\}$	
		$\sigma \neq \tau$	
$N_{\text{gap}} \in \{1, 2\}$	$N_+ = 1$	$\sigma - \tau \equiv \pm 1 \pmod{6}$	R. Vd1 two-chart replacement; route by N'_{gap} .
		$(o(T_\tau), n(T_\tau)) = (1, 1)$	
		$M_\sigma \in T_\tau$	
		$N_E(T_C) = 2$	
		$T_C \cap \{M_i\}_{i=0}^5 = \{M_0\}$	

The short names in the last column are navigation aids. The exact hypotheses and witnesses below are the mathematical interfaces.

5.3. Complementary gap and CE2 short ray.

Lemma 5.5 (One-gap common-pair adapter). *Under the exact row-A hypotheses, rotate the actual gap to*

$$J_0 = X_0([\ell, r]), \quad \ell = B_0, \quad r = 1 - A_1,$$

and put

$$p := 1 - r, \quad q := \ell.$$

Then

$$\forall i \in \mathbb{Z}/6\mathbb{Z}, \quad (p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i).$$

Let

$$0 < p, q < 1, \quad p + q \leq 1, \quad c_* := c_{\max}(p, q),$$

and put

$$J(p, q) := X_0([q, 1 - p]), \quad \mathcal{D}(p, q) := \mathbb{B}(O, h(1 - c_*)).$$

Theorem 5.6 (Complementary-gap enclosure).

$$\Lambda(\mathcal{D}(p, q) \cup J(p, q)) \geq 1.$$

The conclusion includes $p + q = 1$, when $J(p, q)$ is a singleton.

In the six case figures below, orange marks actual gaps, purple marks sampled finite-enclosure points, and the dashed red line is an enclosing triangle. The shading and the displayed triangle are illustrative, not proof objects.

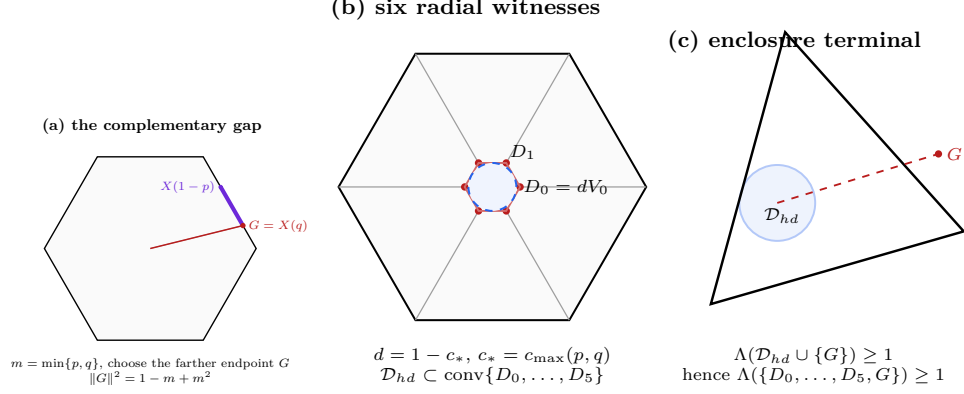
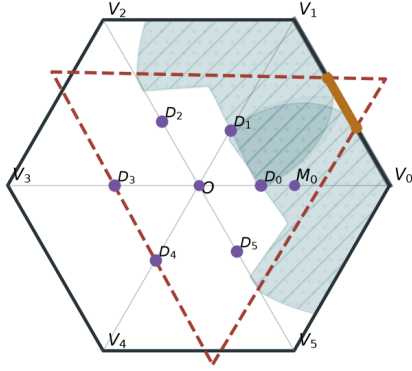
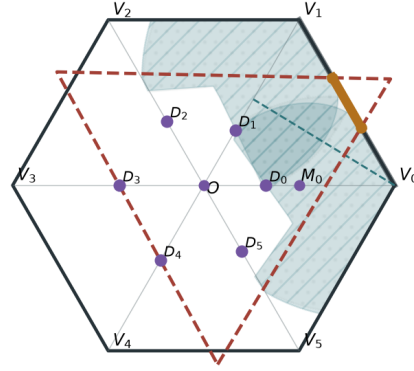


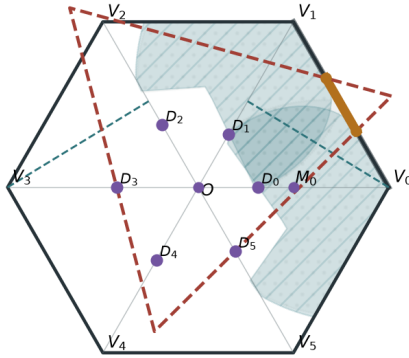
FIGURE 8. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p, q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1 - c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.



All Vd0.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 0)$



One T3-like role.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 1)$



Two T3-like roles.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 2)$

FIGURE 9. Case A: sampled complementary-gap data.

In the CE2 two-gap normalization, define the two positive C-trace endpoints geometrically by

$$p := \min\{x \in [0, 1] : X_5(x) \in T_C\},$$

$$q := 1 - \max\{x \in [0, 1] : X_0(x) \in T_C\},$$

and put

$$c_* := c_{\max}(p, q), \quad D_2 := (1 - c_*)V_2, \quad D_4 := (1 - c_*)V_4.$$

Lemma 5.7 (Two-gap common-pair adapter). *Under the exact row-B hypotheses,*

$$\forall i \in \{1, 2, 3, 4, 5\}, \quad (p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i).$$

For $j \in \mathbb{Z}/6\mathbb{Z}$, if $(o(T_j), n(T_j)) = (2, 1)$ and $\mathcal{H}^1(T_j \cap (r_2 \cup r_4)) > 0$, then

$$A_j + B_j \leq 1, \quad \max\{C_+(A_j, B_j), C_-(A_j, B_j)\} \leq c_*.$$

Theorem 5.8 (CE2 short-ray obstruction). *Assume*

$$N_E(T_C) = 2, \quad N_{\text{gap}} = 2, \quad N_+ \in \{0, 1\}, \quad d = 0, \quad N_+ + t \leq 2.$$

If every $i \in \{1, 2, 3, 4, 5\}$ has either

$$p \leq A_i, \quad q \leq B_i,$$

or

$$q \leq A_i, \quad p \leq B_i,$$

then

$$D_2, D_4 \in U_C, \quad \{D_2, D_4\} \not\subseteq T_C.$$

(a) two CE2 gaps

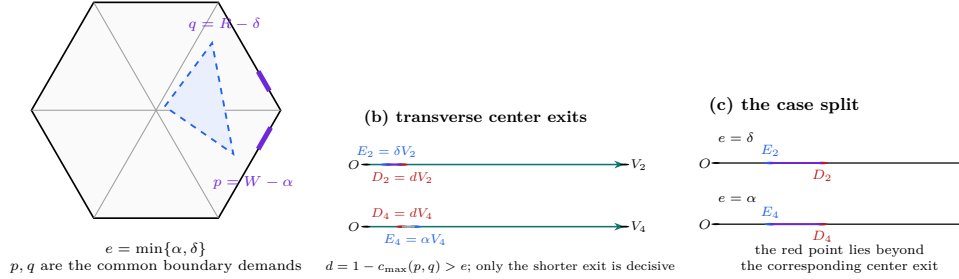
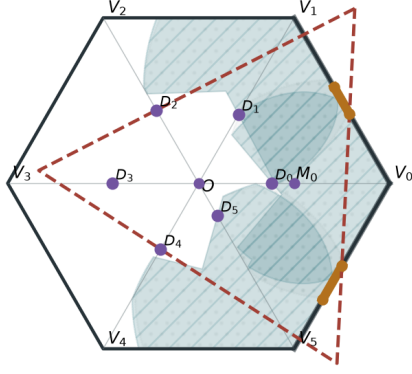
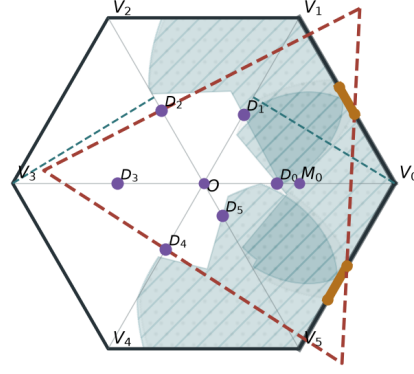


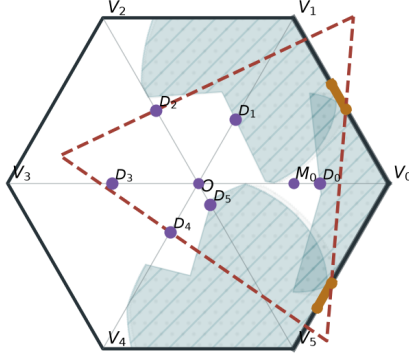
FIGURE 10. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the shorter center residual $e = \min\{\alpha, \delta\}$. The theorem proves $d = 1 - c_{\max}(p, q) > e$. Consequently the radial witness on the shorter of r_2 and r_4 lies strictly beyond the corresponding center exit and is uncovered. The displayed numerical ordering is illustrative; the labelled case split is exact.



All Vd0.
 $N_{\text{gap}} = 2$
 $N_E(T_C) = 2$
 $N_+ = 0$
 $(d, t) = (0, 0)$



Two T3-like roles.
 $N_{\text{gap}} = 2$
 $N_E(T_C) = 2$
 $N_+ = 0$
 $(d, t) = (0, 2)$



One supercritical role.
 $N_{\text{gap}} = 2$
 $N_E(T_C) = 2$
 $N_+ = 1$
 $(d, t) = (0, 0)$

FIGURE 11. Case B: sampled CE2 short-ray data.

Proposition 5.9 (The common-pair nonzero-gap closures). *Every row A or B of table 2 is impossible.*

5.4. **The transverse seven-point witness.** Normalize row C so that

$$J = J_0 = X_0([\ell, r]), \quad \ell := B_0, \quad r := 1 - A_1, \quad 0 < \ell \leq r < 1.$$

For the actual radial reaches define

$$P_i := (1 - C_i)V_i,$$

and define exactly

$$K_{\text{tr}} := \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

Lemma 5.10 (The transverse witness is center-forced). *Under the exact row- C hypotheses,*

$$K_{\text{tr}} \subset U_C.$$

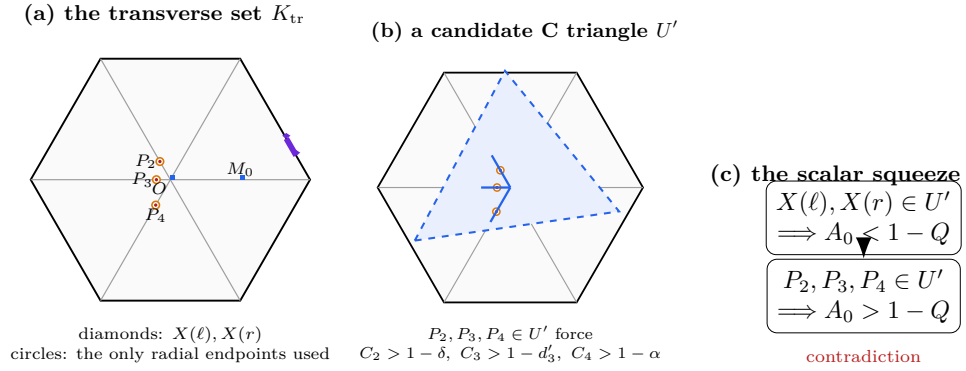


FIGURE 12. The simplified one-gap all-Vd0 enclosure certificate. Only the actual gap endpoints and the three transverse radial endpoints are needed. The gap gives the diameter upper squeeze at the supercritical role; the three radial endpoints supply the CE1/CE2 transverse return. The former points P_0, P_1, P_5 are unnecessary. Lengths are schematic.

Proposition 5.11 (CE2 transverse threshold terminal). *For an open unit equilateral triangle U_C , put $T_C := \overline{U_C}$. Then*

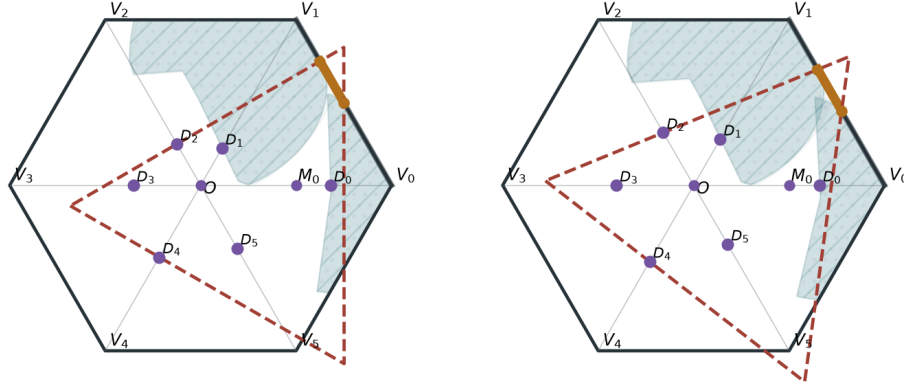
$$N_E(T_C) = 2 \implies K_{\text{tr}} \not\subset U_C.$$

Proposition 5.12 (CE1 three-transverse return). *For an open unit equilateral triangle U_C , put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 1 \implies K_{\text{tr}} \not\subset U_C.$$

Proposition 5.13 (Transverse seven-point enclosure). *Under the exact row- C hypotheses,*

$$\Lambda(K_{\text{tr}}) \geq 1.$$

**CE1 terminal context.**

$$\begin{aligned}
 N_{\text{gap}} &= 1 \\
 N_+ &= 1 \\
 (d, t) &= (0, 0) \\
 N_E(T_C) &= 1
 \end{aligned}$$

CE2 terminal context.

$$\begin{aligned}
 N_{\text{gap}} &= 1 \\
 N_+ &= 1 \\
 (d, t) &= (0, 0) \\
 N_E(T_C) &= 2
 \end{aligned}$$

FIGURE 13. Case C: sampled transverse-witness data.

5.5. Supported tails and radial separation. For $0 \leq c < 1/2$, define

$$M_c^{\text{sup}} := \sup\{M_c(a) : 0 \leq a \leq 1, M_c(a) > 1 - a\},$$

and set

$$M_{1/2}^{\text{sup}} := \frac{1}{2}.$$

The endpoint value is the continuous extension, not the supremum of an empty strict feasible set.

Both supported-tail adapters use an actual interval

$$\begin{aligned}
 T_0 \cap r_1 &= \{V_1 + x(O - V_1) : c \leq x \leq u\}, \\
 0 < c &\leq \frac{1}{2} \leq u < 1, \quad \varepsilon := 1 - u, \quad C_1 \geq c.
 \end{aligned}$$

With $M := M_c^{\text{sup}}$, their common exact conclusion is

$$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M - B_1 > 4.$$

Proposition 5.14 (T3-like supported-tail terminal). *Assume*

$$\begin{aligned}
 N_{\text{gap}} &\in \{1, 2\}, \quad N_+ = 1, \quad (d, t) = (0, 1), \quad N_E(T_C) \in \{1, 2\}, \\
 T_C \cap \{M_0, \dots, M_5\} &= \{M_0\}, \quad (o(T_0), n(T_0)) = (2, 1), \\
 (o(T_i), n(T_i)) &\in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5).
 \end{aligned}$$

Then the skeleton is not covered.

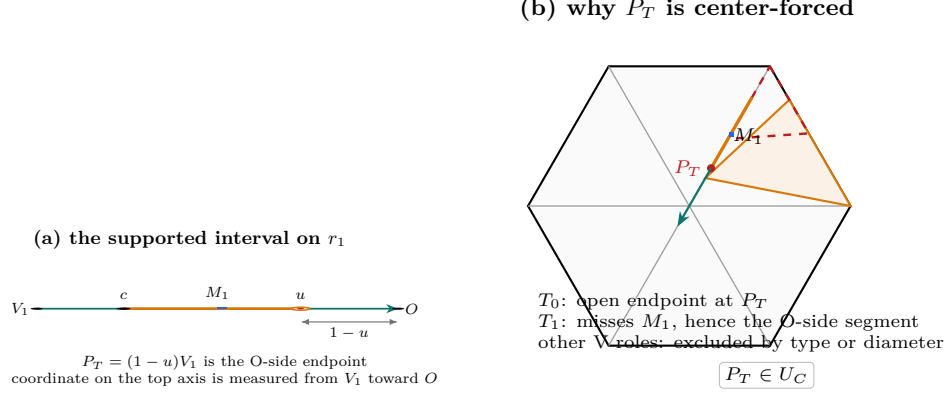


FIGURE 14. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1-u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

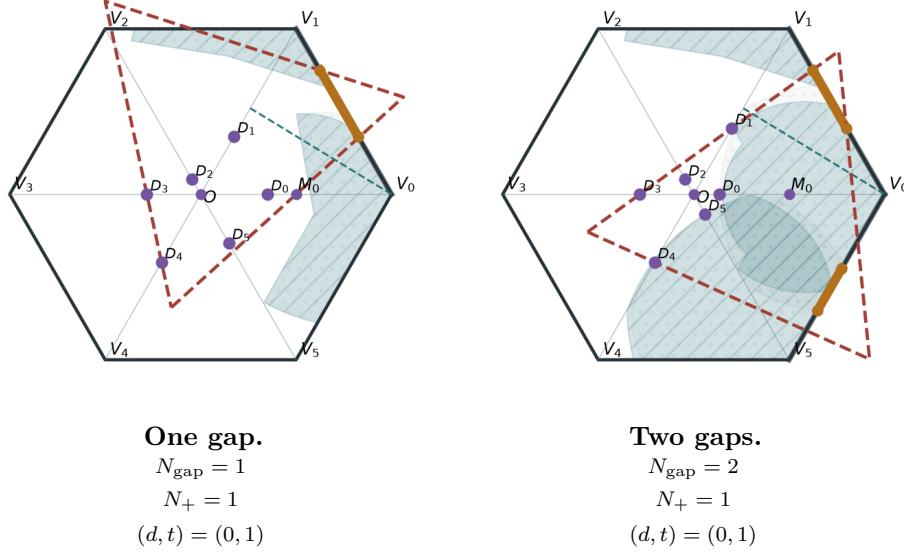


FIGURE 15. Case D_T : sampled T3-like supported-tail data.

For radial separation along r_i , define

$$d_i^C := \max\{x \in [0, 1] : xV_i \in T_C\},$$

$$c_{\text{loc}} := \sup\{s \in [0, 1] : V_i + s(O - V_i) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \leq x \leq d_i^C\},$$

$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \leq x \leq 1\}.$$

Thus the exact strict inequality

$$c_{\text{loc}} < 1 - d_i^C$$

leaves the nonempty interval

$$\{xV_i : d_i^C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

For the nonadjacent adapter, if $I \subseteq [0, 1]$ is empty or a closed interval, define

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup I\}.$$

Put

$$\begin{aligned} I_R &:= \{x \in [0, 1] : X_0(x) \in T_C\}, & I_L &:= \{x \in [0, 1] : X_5(1-x) \in T_C\}, \\ \rho_R &:= \mathcal{R}_{I_R}(B_0), & \rho_L &:= \mathcal{R}_{I_L}(A_0), & \rho &:= \min\{\rho_R, \rho_L\}, & D_\tau &:= \rho V_\tau. \end{aligned}$$

Appendix D proves the exact separation inequalities for both adapters.

5.6. The Vd placement router.

Proposition 5.15 (Direct one-Vd placement assembly). *Assume*

$$\begin{aligned} N_E(T_C) &= 2, & N_{\text{gap}} &\in \{1, 2\}, & N_+ &= 1, & (d, t) &= (1, 0), \\ T_C \cap \{M_0, \dots, M_5\} &= \{M_0\}. \end{aligned}$$

Then the skeleton is not covered.

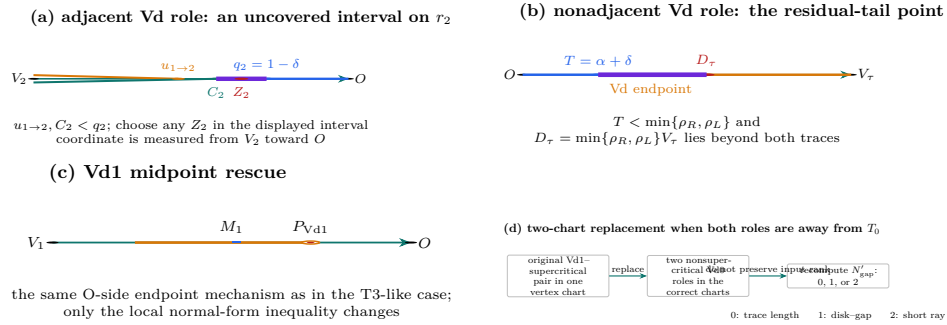
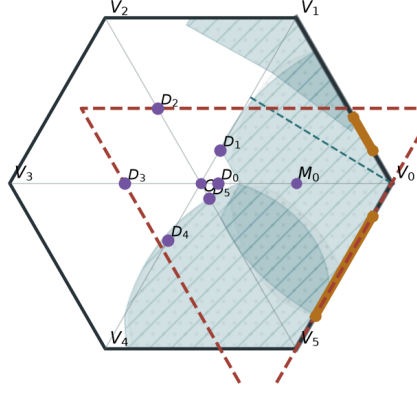
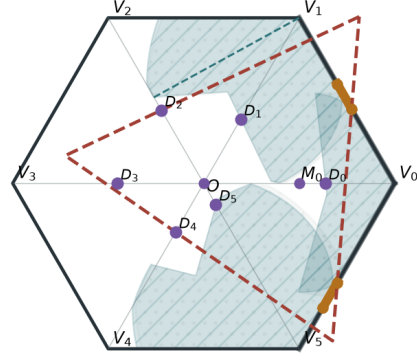


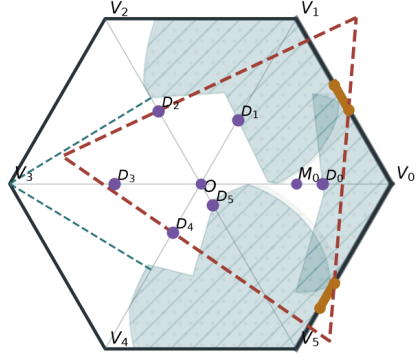
FIGURE 16. The four geometric terminals in the one-Vd placement assembly. In the adjacent placement the radial coordinate is measured from V_2 toward O ; the two local V endpoints lie before the center interval entry $q_2 = 1 - \delta$, leaving a literal uncovered interval on r_2 . The nonadjacent placement produces the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ beyond both the Vd and center traces; the Vd1 midpoint rescue uses the same O-side endpoint construction as the T3-like case; and the remaining placement is reduced by a two-chart replacement followed by recomputation of the output gap rank. Lengths are schematic; all labelled order relations are the proof relations.

**Vd1 supported tail.**

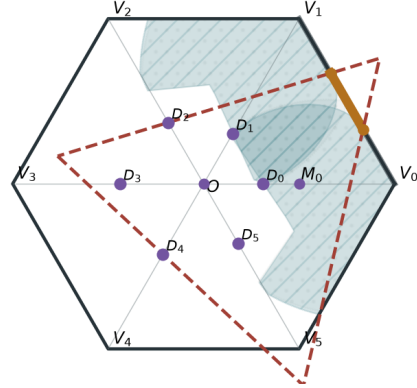
$$\begin{aligned}
 N_{\text{gap}} &= 2 \\
 N_+ &= 1 \\
 (d, t) &= (1, 0) \\
 (\sigma, \tau) &= (1, 0) \\
 (o(T_0), n(T_0)) &= (1, 1) \\
 A_0 + B_0 &< 1/2
 \end{aligned}$$

**Adjacent separation.**

$$\begin{aligned}
 N_{\text{gap}} &= 2 \\
 N_+ &= 1 \\
 (d, t) &= (1, 0) \\
 (\sigma, \tau) &= (0, 1)
 \end{aligned}$$

**Nonadjacent separation.**

$$\begin{aligned}
 N_{\text{gap}} &= 2 \\
 N_+ &= 1 \\
 (d, t) &= (1, 0) \\
 (\sigma, \tau) &= (0, 3)
 \end{aligned}$$

**Replacement output $R \rightarrow A$.**

$$\begin{aligned}
 N'_{\text{gap}} &= 1 \\
 A(T'_i) + B(T'_i) &\leq 1 \quad (i \in \mathbb{Z}/6\mathbb{Z}) \\
 (o(T'_i), n(T'_i)) &\in \{(1, 0), (2, 0)\} \quad (i \in \mathbb{Z}/6\mathbb{Z})
 \end{aligned}$$

FIGURE 17. Cases D_V , E_a , E_n , and R: sampled router data.

Proposition 5.16 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of table 1 assigned to finite enclosure is impossible.*

5.7. The zero-gap nine-point obstruction. Assume

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$

Rotate the unique supercritical role to index 4. The exact-one handoff calculation theorem E.2 gives

$$(14) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

Define

$$(15) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a.$$

Then

$$(16) \quad \begin{aligned} 0 < a, b < 1, \quad a &= a_4 < A_4, \quad b = b_4 < B_4, \\ a + b &> 1, \quad a^2 + ab + b^2 < 1, \\ a_i &\geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p, q), \quad r_* := h(1 - c_*), \quad P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The radial forcing calculation gives

$$(17) \quad \mathbb{B}(O, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

Define ℓ_- and ℓ_+ as the corresponding maximal nondegenerate line segments of the interior frontier:

$$\begin{aligned} \{\ell_-\} &:= \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_- \neq \emptyset, \\ \nexists X', Y' [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}, \\ \{\ell_+\} &:= \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_+ \neq \emptyset, \\ \nexists X', Y' [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}. \end{aligned}$$

The three frontier points are the singleton intersections

$$\{Q_0\} := \ell_- \cap \ell_+, \quad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Their exact construction and exclusions in theorems E.1, E.4 and E.5 give

$$(18) \quad Q_-, Q_0, Q_+ \in U_C.$$

Define the public witness set

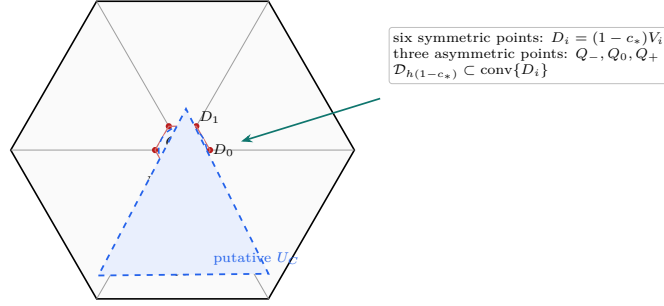
$$(19) \quad K_{\text{wit}}(a, b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

Theorem 5.17 (Witness enclosure). *Every pair (a, b) satisfying equation (16) satisfies*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

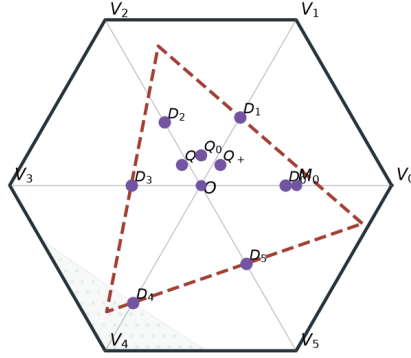
Theorem 5.18 (Zero-gap obstruction). *There is no hypothetical cover with*

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$



$$K_{\text{wit}} = \mathcal{D}_{h(1-c_*)} \cup \{Q_-, Q_0, Q_+\} \subset U_C \text{ under a hypothetical cover}$$

FIGURE 18. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

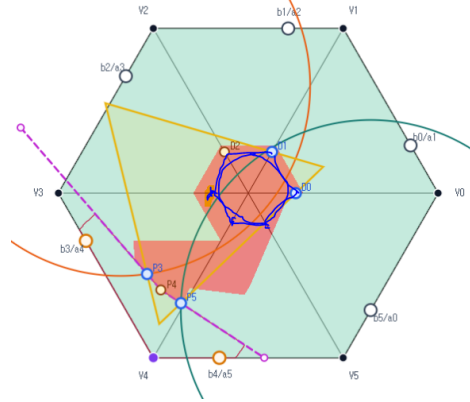


Trace-exact witness.

$$N_{\text{gap}} = 0$$

$$N_+ = 1$$

$$(d, t) = (0, 0)$$



Core configuration.

$$N_{\text{gap}} = 0$$

$$N_+ = 1$$

$$(d, t) = (0, 0)$$

FIGURE 19. Case F: sampled zero-gap obstruction data.

Proposition 5.19 (Cases excluded by the nine-point obstruction). *Every $N_{\text{gap}} = 0$, $N_+ = 1$, $(d, t) = (0, 0)$ row of table 1 is impossible, independently of the center class.*

6. COMPLETION OF THE PROOF

Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER OPTIMIZATION

This appendix solves the coordinate and compactness problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. Open–closed shrinking feasibility.

Lemma A.1 (Uniform shrinking margin). *If seven open unit equilateral triangles cover H , then, for some $\varepsilon > 0$, seven closed equilateral triangles of side $1 - \varepsilon$ also cover H .*

A.2. T3-like trace-majorant feasibility. The calculation uses the local wedge and the two orientation charts developed in the exact-trace module below. It is placed here beside the body warning that the output is a closed majorant, not a replacement open triangle.

Lemma A.2 (T3-like translation calculation). *The closed-trace majorant in Proposition 2.7 exists. In the calculation chart, every original T3-like triangle also satisfies $A_i + B_i \leq 1$.*

A.3. Strict-handoff feasibility.

Lemma A.3 (Cyclic strict-handoff selection). *Under the hypotheses of Proposition 2.11, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

A.4. Shared equilateral-enclosure optimization. Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition A.4 (Equilateral enclosure gauge). *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$(20) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a polygonal hull with at least two distinct points, a minimizing normal can be chosen perpendicular to a hull edge.

A.4.1. Finite caliper criterion. Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$(21) \quad W_F(N) := \sum_{k=0}^2 \max_{z \in F} \langle z, R^k N \rangle.$$

Theorem A.5 (Finite caliper certificate). *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(22) \quad W_F(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length. For $F_x = \{O, A, B, x\}$, every clause is a finite polynomial clause of degree at most four in the coordinates of A, B, x ; at most $12 \cdot 4^3$ clauses are needed.

A.5. Center and V triangle incidence feasibility.

Lemma A.6 (Separation of nonadjacent boundary edges). *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Lemma A.7 (Local wedge calculation). *The incidence conclusions of Lemma 2.4 hold.*

A.6. Exact-trace normalization. Set

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

In Type I write

$$(23) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(24) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both charts

$$(25) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(26) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Lemma A.8 (Raw $(3, 0)$ trace-normalization calculation). *The translation asserted in Proposition 2.5 exists and preserves both the open and closed traces in H .*

A.7. The local admissible set and radial envelope. Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \text{conv}\{0, au, bv, c(u + v)\}.$$

This is the translate of the body normalization at V_0 , with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$(27) \quad \mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.9 (Exact local admissible set). *Put*

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & m &:= \min\{a, b\}, & M &:= \max\{a, b\}, \\ q &:= s^4 - s^2 + ab, \end{aligned}$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then

$$(28) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where all variables lie in $[0, 1]$ and

$$\begin{aligned} \mathcal{A}_L &:= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &:= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &:= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Then

$$(29) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ \frac{C_L(m)}{2M}, & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ \frac{1 + \sqrt{4s^2 - 3}}{2M}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.8. Midpoint caliper optimization.

Lemma A.10 (Four-point self-midpoint caliper). *If $a + b > 1$, then*

$$\Lambda\left(\text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.12.

A.9. Signed-center optimization. The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.11 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E), \end{aligned}$$

and nonnegative slacks α, δ , with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (30) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned}$$

Define

$$(31) \quad \Delta_R := P - \alpha - W\delta, \quad \Delta_L := P - R\alpha - \delta.$$

Then

$$(32) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(33) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalized trace requires $\Delta_R > 0$, and

$$(34) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

$$\begin{aligned} (35) \quad d_0^C &= E - \alpha - \delta, \quad d_1^C = \frac{\delta}{R}, \quad d_2^C = \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, \quad d_4^C = \alpha, \quad d_5^C = \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(36) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Lemma A.12 (CE2 total slack). *In the CE2 sign range, with $T := \alpha + \delta$,*

$$(37) \quad T < \frac{\eta}{E}.$$

Corollary A.13 (Common center-boundary formula). *The full center-boundary contribution is*

$$(38) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Lemma A.14 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \leq M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. Conditional skeleton caps.

Lemma B.1 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.1, one has $L_S(T_C) < 3/2$.*

Lemma B.2 (No-support skeleton calculation). *The hypotheses of Lemma 3.3 imply $L_S(T_i) \leq 2$.*

For the remaining cap, reflect the local directions if necessary so that $0 \leq A_i \leq B_i \leq 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$(39) \quad \begin{aligned} A_i^2 + A_i B_i + B_i^2 &\leq 1, \\ C_i &\leq \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_i B_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\leq 0, \end{aligned}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.9. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.3 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Lemma B.4 (Common skeleton-budget calculation). *Under the hypotheses of Theorem 3.6, the total skeleton trace of the seven triangles is strictly less than 12.*

B.2. Vd1 and Vd2 corner optimization. Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition B.5 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A_0, B_0 be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$(40) \quad T = \left\{ (x, y) : \begin{aligned} x - (t+1)y &\leq A_0, \\ ty - (t+1)x &\leq tB_0, \\ tx + y &\leq d - A_0 - tB_0 \end{aligned} \right\}.$$

The raw traces, parametrized from the indicated outer vertex, are

$$(41) \quad \begin{array}{l} r_0 \\ r_1 \\ r_5 \end{array} \left| \begin{array}{l} 0 \leq s \leq \frac{d - A_0 - tB_0}{t+1} \\ \frac{t(1-B_0)}{t+1} \leq s \leq \frac{d - A_0 - tB_0 - 1}{t} \\ \frac{1-A_0}{t+1} \leq s \leq d - A_0 - tB_0 - t. \end{array} \right.$$

Intersecting with $[0, 1]$ gives the actual traces, and

$$(42) \quad A_0 + B_0 < \frac{1}{2}.$$

Lemma B.6 (Vd2 adjacent-midpoint cap). *If an original open V triangle has Vd2 closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

B.3. Boundary contributions.

Lemma B.7 (Boundary-trace table calculation). *All identities and bounds in Theorem 3.7 hold.*

B.4. Common perimeter budgets.

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta := \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has center contribution $L_C := L_{\partial H}(T_C)$, exactly n supercritical Vd0 triangles, and $m \geq 1$ distinct nonsupercritical V triangles $T_{i_1}, \dots, T_{i_m}$ such that

$$L_{\partial H}(T_{i_j}) < q_j < 1 \quad (1 \leq j \leq m).$$

Suppose also that each of the remaining $6 - n - m$ V triangles contributes at most 1. If

$$(43) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Lemma B.9 (Small center slacks in the one-Vd branch). *Assume there is exactly one supercritical Vd0 triangle, exactly one Vd1 or Vd2 triangle, and four nonsupercritical Vd0 triangles. Every candidate surviving the perimeter budget is CE2 and satisfies*

$$(44) \quad \alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

Lemma B.10 (CE2 endpoints dominate total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T := \alpha + \delta, \quad x := \frac{\eta + T}{W}, \quad y := \frac{\eta + T}{R}.$$

Then

$$(45) \quad T < \frac{1}{2} \min\{x, y\}.$$

Lemma B.11 (Perimeter-budget substitutions). *Items (1)–(5) of Corollary 3.8 are impossible.*

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix evaluates the three area optimizations used in section 4. The body fixes the feasible triangles, the loss functional, and the strict cyclic handoffs; here we introduce coordinates only to solve those problems.

C.1. Wedge chart.

Lemma C.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

C.2. Local square-loss optimization.

Lemma C.2 (Solved local square-loss optimization). *For the feasible family $\mathcal{T}(a, b)$ of section 4, the minimum loss satisfies equations (10) and (11).*

C.3. T3-like limiting optimization.

Lemma C.3 (Solved T3-like loss optimization). *Under the hypotheses of theorem 4.2, the actual V triangle satisfies equation (13).*

C.4. Cyclic loss optimization. The variables ξ_i, a_i, b_i are the strict handoffs already fixed in section 4; in particular, no endpoint selection is relaxed here.

Lemma C.4 (Solved cyclic area-loss optimization). *Both implications in theorem 4.3 hold.*

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix solves the point, radial, disk, and interval calculations used by the nonzero-gap rows of table 2. The geometric witnesses and their case predicates remain in the body.

D.1. Exact gap endpoints.

Lemma D.1 (Actual open-trace endpoints). *For every actual gap J_i ,*

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

The formula includes the singleton case $B_i = 1 - A_{i+1}$.

D.2. Common-pair reach adapters.

Lemma D.2 (Solved one-gap common-pair adapter). *The reaches defined from row A in theorem 5.5 satisfy its displayed coordinatewise alternatives.*

Lemma D.3 (Solved two-gap common-pair adapter). *The reaches defined from row B in theorem 5.7 satisfy its displayed coordinatewise alternatives.*

D.3. Vd1 two-chart replacement. The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then recomputes the output gap rank from the new open traces.

Lemma D.4 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Proposition D.5 (Vd1 two-chart replacement and router). *Assume the normalized replacement hypotheses in the proof of theorem 5.15: T_0 is Vd1, T_1 is the unique supercritical Vd0 role, $M_1 \in T_0$, and only T_0, T_1 have positive trace on r_0 or r_1 . Then there are two open unit equilateral triangles U'_0, U'_1 that preserve the covered skeleton, make all six V roles nonsupercritical Vd0, and route by the recomputed $N'_{\text{gap}} \in \{0, 1, 2\}$.*

D.4. Neighboring-arm optimization. Normalize at V_0 . Let $A_a \in e_{5,0}$ and $B_b \in e_{0,1}$ have reaches a, b , and put

$$X_c = (1 - c)V_1 \in r_1.$$

Thus $C_+(a, b)$ is the maximal c for which V_0, A_a, B_b, X_c lie in a closed unit equilateral triangle, and $C_-(a, b) = C_+(b, a)$.

For $0 \leq a \leq 1/2$, let $p(a)$ be the unique root in $[a, 1]$ of

$$(46) \quad p^3 - (a + 2)p^2 + 2(a + 1)p - 1 = 0,$$

and define

$$\sigma(a) = 1 - p(a), \quad \tau(a) = 1 - a - (p(a) - a)(1 - p(a)).$$

Proposition D.6 (Neighboring-ray formula).

$$(47) \quad C_+(a, b) = \begin{cases} \text{undefined}, & a + b > 1, \\ 1 - b, & a + b \leq 1, \ a > 1/2, \\ 1 - b, & 0 \leq a \leq 1/2, \ 0 \leq b \leq \sigma(a), \\ p(a), & 0 \leq a \leq 1/2, \ \sigma(a) \leq b \leq \tau(a), \\ a + \frac{1}{2} - \sqrt{(a + b)^2 - \frac{3}{4}}, & 0 \leq a \leq 1/2, \ \tau(a) < b \leq 1 - a. \end{cases}$$

At $b = \tau(a)$ the plateau value $p(a)$ is selected.

A consequence used repeatedly below is

$$\boxed{C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\}} \quad (p + q \leq 1).$$

Indeed this is immediate on the linear branch; on the plateau branch the cubic gives $p(a) \leq 1 - a$, and the radical branch is no larger than its selected plateau at the transition.

Lemma D.7 (Type-aware radial calculation). *The points $D_i = (1 - \Gamma_i)V_i$ of theorem 5.2 are missed by every open V role.*

Lemma D.8 (Solved common-pair comparison). *The inequalities in theorem 5.3 hold.*

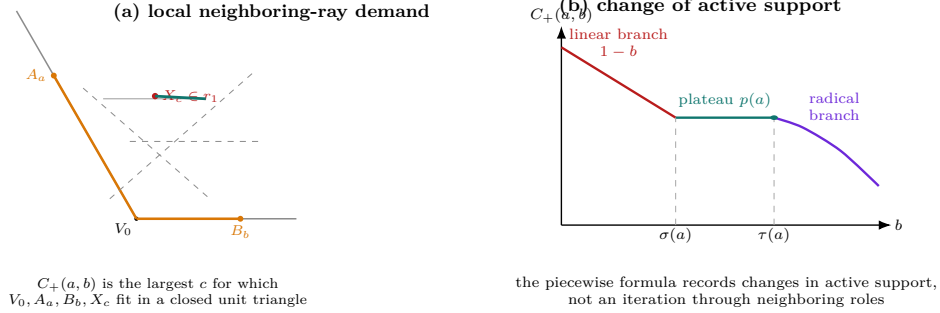


FIGURE 20. Geometry behind the neighboring-ray capacity $C_+(a, b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1 . As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

D.5. Disk and complementary-gap optimization. Put $\mathcal{D}_\eta := \mathbb{B}(O, \eta)$.

Lemma D.9 (Disk-plus-point formula). *If $\|G\| = \rho$, then*

$$\Lambda(\mathcal{D}_\eta \cup \{G\}) = \begin{cases} 3\eta/h, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}, & \rho > 2\eta. \end{cases}$$

Lemma D.10 (Solved complementary-gap optimization). *The set $\mathcal{D}(p, q) \cup J(p, q)$ of theorem 5.6 has enclosure number at least one.*

D.6. CE2 short-ray optimization.

Lemma D.11 (Solved CE2 short-ray optimization). *Under the hypotheses of theorem 5.8,*

$$c_{\max}(p, q) < 1 - e, \quad e := \min\{\alpha, \delta\},$$

and consequently $\{D_2, D_4\} \not\subseteq T_C$.

D.7. High-radial and selected-chord calculations. The zero-radial diameter envelope is

$$(48) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}.$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$(49) \quad \epsilon(e) = \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3} \right).$$

It satisfies

$$(50) \quad e < \epsilon(e), \quad \epsilon(e) < \frac{2e}{1-2e},$$

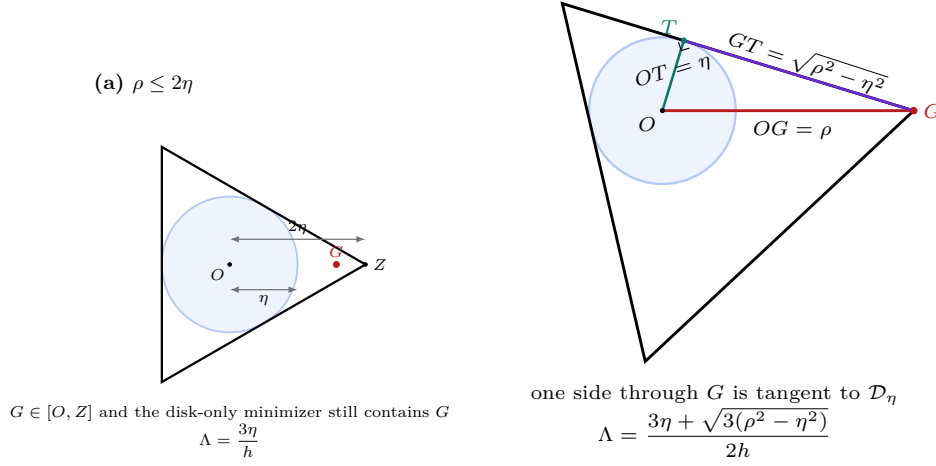
(b) $\rho > 2\eta$ (a) $\rho \leq 2\eta$ 

FIGURE 21. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

and, for $e \leq 1/8$,

$$(51) \quad \epsilon(e) \leq 2e + 5e^2.$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.12 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B . If*

$$C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$\boxed{B \leq \epsilon(e)}.$$

For $0 \leq c \leq 1/2$, the strict-supercritical outgoing supremum is

$$M_c^{\text{sup}} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained. Thus an actual strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below M_c^{sup} .

Lemma D.13 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2}.$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$q \geq p + \lambda(p - e)$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$\lambda > 1 - 4e \quad (e = \alpha), \quad \lambda > 1 - 5e \quad (e = \alpha/R).$$

D.8. Exact direct output formulas. The preceding proofs refer to three local inequalities. We derive them here from Proposition A.9. This replaces the former presentation by iterated reach maps.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c , let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is an outgoing capacity for one triangle; it is never composed in the proof.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) = c - \ell(c).$$

Also set

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a, \\ Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition D.14 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

D.8.1. *Vd corner margins.* In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b , let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\},$$

and

$$u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \quad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

D.9. CE1 reverse-path calculation. The CE1 part of the seven-point enclosure theorem uses five boundary handoffs and the three transverse radial reaches at T_4, T_3, T_2 .

Proposition D.15 (CE1 three-transverse return certificate). *Use the signed CE1 center variables*

$$\begin{aligned} 0 < R < 1, \quad w &= 1 - R, \quad E = \sqrt{1 - Rw}, \\ \eta &= 1 - E, \quad P = E(1 - E), \end{aligned}$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$(52) \quad A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$(53) \quad X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let T_0 be the unique supercritical role and let $T_1, \dots, T_5$ be nonsupercritical Vd0 roles. Assume

$$\begin{aligned} A_1 &\geq X, \\ B_i + A_{i+1} &\geq 1 \quad (1 \leq i \leq 4), \quad B_5 + A_0 \geq 1, \end{aligned}$$

and

$$(54) \quad C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

Then

$$(55) \quad \boxed{A_0 > 1 - h_0.}$$

CE1 reverse path: from a deficient terminal to the final contradiction

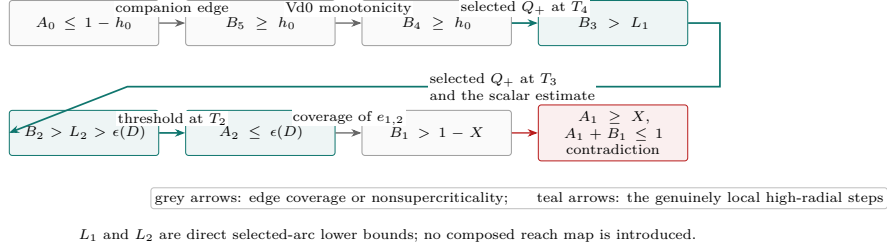


FIGURE 22. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0 , moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

D.10. Transverse seven-point calculation.

Lemma D.16 (Endpoint upper squeeze). *If an open unit triangle U' contains K_{tr} , use its local signed-center variables and put*

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

Then its incoming reach at V_0 satisfies

$$A_0(U') < 1 - Q.$$

Lemma D.17 (Solved CE2 transverse threshold). *For an open unit equilateral triangle U_C , put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 2 \implies K_{\text{tr}} \not\subseteq U_C.$$

D.11. Disk, one-third, and supported-tail optimizations.

Proposition D.18 (Disk-finite-set calipers). *Let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Lemma D.19 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$(56) \quad \boxed{c_{\max}(M, m) < 1 - \frac{m}{3}}.$$

Lemma D.20 (Rescuer endpoint and boundary-tail budget). *Suppose a special role at V_0 has boundary endpoint a on $e_{5,0}$ and supported interval $[c, u] \subset r_1$, measured from V_1 toward O . Put*

$$\varepsilon = 1 - u, \quad M = M_c^{\sup}.$$

Assume

$$(57) \quad \begin{aligned} \varepsilon V_1 &\in U_C, & a + \varepsilon &\leq 1, \\ a &\leq 1 - M, & \frac{a}{a + \varepsilon} &\leq 1 - M. \end{aligned}$$

If T_1 is uniquely supercritical with own-radial reach at least c and T_2, T_3, T_4, T_5 are non-supercritical on the center-free four-edge path, then the skeleton is not covered.

D.12. T3-like supported-tail adapter.

Lemma D.21 (Solved T3-like supported-tail adapter). *The exact hypotheses of theorem 5.14 imply the rescuer-tail hypotheses in theorem D.20.*

D.13. One-Vd radial-separation calculations.

Lemma D.22 (Solved adjacent Vd separation). *Under the adjacent placement*

$$\sigma = 0, \quad \tau \in \{1, 5\},$$

the radial-separation inequality $c_{\text{loc}} < 1 - d_2^C$ holds on the routed arm r_2 .

Lemma D.23 (Solved nonadjacent Vd separation). *Under the nonadjacent placement*

$$\sigma = 0, \quad \tau \in \{2, 3, 4\},$$

the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ lies beyond both the C -triangle exit and the actual Vd radial endpoint.

Lemma D.24 (Solved Vd1 supported-tail adapter). *When the unique Vd1 role is based at T_0 , its actual corner data imply the two strict-supercritical envelope inequalities required by theorem D.20.*

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix supplies the exact frontier, radial, pointwise, Newton, and support-arc calculations for the zero-gap nine-point obstruction. It uses the public parameters and witness $K_{\text{wit}}(a, b)$ from section 5.7. The symbols A, B, C introduced below are appendix-only Newton inner points required by the exact certificate; they do not replace the public frontier points Q_-, Q_0, Q_+ .

The finite caliper criterion used here is the shared result theorem A.5 in Appendix A. It is not rederived in this appendix.

E.1. Corner-clipped AB -unions. Apply the rigid motion that sends the supercritical corner V_4 to the local origin and its two incident edge directions to e_A, e_B . Within this calculation O, A, B, C denote the local origin, anchors, and corner cone; the feasible union keeps the public symbol $\mathcal{R}_{AB}(a, b)$.

Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

(58)

$$\mathcal{R}_{AB}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem A.5 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (58) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(59) \quad \mathcal{R}_{AB}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\mathcal{D}_A = \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\},$$

$$\mathcal{D}_B = \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\},$$

$$\mathcal{H}_A = \{(u, v) : \alpha(u - a) + \beta v \leq 0\},$$

$$\mathcal{H}_B = \{(u, v) : \gamma u + \delta(v - b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

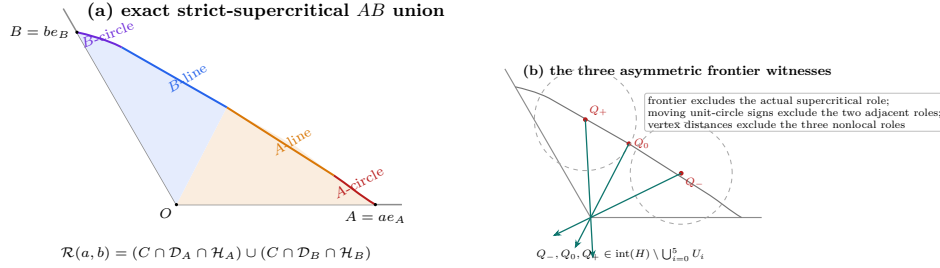


FIGURE 23. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

E.2. Exact-one handoffs and common reach bounds.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(60) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

E.3. Six directly forced radial points.

Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (60) gives

$$(61) \quad pq > (1 - s)(2 - s), \quad m > 1 - s, \quad s > 1 - m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.9. On the present subcritical domain, equation (29) becomes

$$(62) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(63) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that every $T_i = \overline{U_i}$ is Vd0 and that $U_C, U_0, \dots, U_5$ cover H . Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(64) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem E.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$(65) \quad J = \left(\frac{2b + a - aD}{D+3}, \frac{b + 2a - bD}{D+3} \right).$$

Put

$$E_- = (1 - b, 2 - b), \quad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \quad 1 - a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \quad 1 - b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(66) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (60),*

$$(67) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1 - a, b]$,

$$(68) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1 - b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$(69) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and by the displayed conversion from (u, v) to the plane.

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas E.3 and E.5 therefore force the compact set

$$(70) \quad K_{\text{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

E.4. Tangent reduction and exact mixed overlaps. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (66), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_- \right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \text{conv}(K_{\text{wit}})$.

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1, \langle X, n \rangle + r \geq h\}.$$

E.5. Ray order and four overlaps.

Proposition E.7 (Technical four-overlap verification). *Assume (60) and $c_* > 2/3$. Then*

$$\|A\|, \|C\| > \eta,$$

the rays A, B, C, W, RA occur in that cyclic order, and the four consecutive cap pairs

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

overlap across their intervening short sectors.

E.6. Newton reduction in the body notation. The containment established by theorem E.6 is recorded under the public routing label

$$(71) \quad \widehat{K} = \text{conv}(\mathbb{B}(O, \eta) \cup \{A, B, C\}) \subseteq \text{conv}(K_{\text{wit}}(a, b)), \quad \Lambda(K_{\text{wit}}(a, b)) \geq \Lambda(\widehat{K}).$$

E.7. Support arcs and the enclosure conclusion. Let R denote counterclockwise rotation through $2\pi/3$. Retain the points A, B, C , the vector $W = C + RA$, the forced disk $\mathcal{D}_\eta$, and the exact witness set K_{wit} constructed above. For uniform notation put

$$P_- = A, \quad P_0 = B, \quad P_+ = C, \quad P_* = W, \quad r_* = \eta,$$

and

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}).$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\},$$

and define the level- h , $h = \sqrt{3}/2$, support arc

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Lemma E.8 (Cyclic support-arc covering). *Let K be compact and put*

$$\Sigma_K = K + RK + R^2K.$$

Suppose Σ_K contains the full R -orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

Assume that the rays

$$P_-, P_0, P_+, P_*, RP_-$$

occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then

$$\Lambda(K) \geq 1.$$

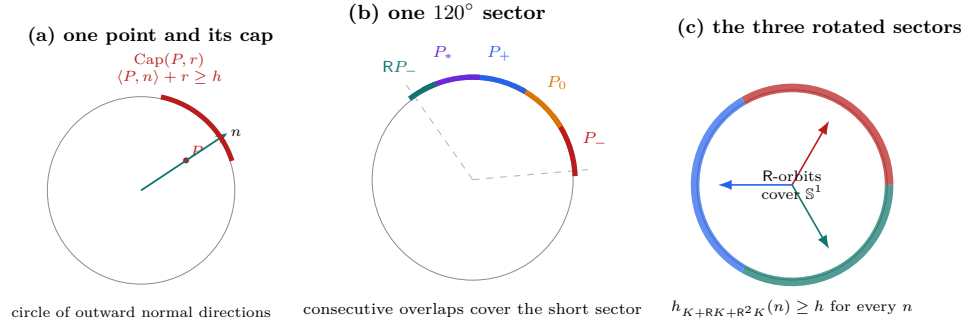


FIGURE 24. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h . The four consecutive overlap checks cover the sector from the ray of P_- to that of RP_- , and the three 120° rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

Lemma E.9 (Solved witness-enclosure optimization). *For every parameter pair in equation (16),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

APPENDIX F. EXACT UNEQUAL-RADIUS SUPPORT-ARC INTERSECTIONS

The two mixed cap overlaps in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of Section E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$(72) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(73) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_$.*

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\begin{aligned} \Xi(U, z) &= 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2, \\ G(U, z) &= 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z. \end{aligned}$$

Thus

$$(74) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (74), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C'\|^2, \quad \nu = \langle A, C' \rangle,$$

$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$(75) \quad h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(76) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the mixed cap intersections). *If $Q_A, Q_C \geq 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + RA, \bar{\eta}) \neq \emptyset,$$

and

$$I(C + RA, \bar{\eta}) \cap I(RA, 2\bar{\eta}) \neq \emptyset.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (74) into (76), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(77) \quad \text{num}(Q_{B,X}) = 32(K+3)^2 \{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. Three global positive-basis certificates. Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\Delta = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U -range	z^2 -range
T	$0 \leq U \leq U_\Delta$	$0 \leq z^2 \leq \mathcal{P}(U)$
L_0	$0 \leq U \leq U_\Delta$	$\mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U)$
L_1	$U_\Delta \leq U \leq U_{\max}$	$0 \leq z^2 \leq \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_\Delta r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_\Delta + (U_{\max} - U_\Delta)r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij} r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact mixed-overlap certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (77) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two mixed cap intersections required by Proposition E.7 hold.*

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
```

```
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.